

Variational Monte Carlo calculations of $A \leq 4$ nuclei with an artificial neural-network correlator ansatz

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The complexity of many-body quantum wave functions is a central aspect of several fields of physics and chemistry where non-perturbative interactions are prominent. Artificial neural networks (ANNs) have proven to be a flexible tool to approximate quantum many-body states in condensed matter and chemistry problems. In this work we introduce a neural-network quantum state ansatz to model the ground-state wave function of light nuclei, and approximately solve the nuclear many-body Schrödinger equation. Using efficient stochastic sampling and optimization schemes, our approach extends pioneering applications of ANNs in the field, which present exponentially-scaling algorithmic complexity. We compute the binding energies and point-nucleon densities of $A \leq 4$ nuclei as emerging from a leading-order pionless effective field theory Hamiltonian. We successfully benchmark the ANN wave function against more conventional parametrizations based on two- and three-body Jastrow functions, and virtually-exact Green's function Monte Carlo results.

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Introduction – The last two decades have witnessed remarkable progress in our understanding of how the structure and dynamics of atomic nuclei emerge from the individual interaction among protons and neutrons. This progress has been primarily driven by the widespread use of nuclear-effective field theories to systematically construct realistic Hamiltonians [1–3], and the concurrent development of nuclear many-body techniques that solve the time-independent Schrödinger equation with controlled approximations [4–8]. The variational Monte Carlo (VMC) and the Green's function Monte Carlo (GFMC) methods are ideally suited to tackle this problem and have been extensively applied to study properties of light nuclei [9]. The calculation of the spin-isospin dependent Jastrow correlations used in the VMC and GFMC scales exponentially with the number of nucleons, limiting the applicability of these methods to relatively small nuclear systems. The auxiliary-field diffusion Monte Carlo [10] (AFDMC) samples the spin and isospin degrees of freedom to treat larger nuclei and infinite nucleonic matter [11, 12], but it can only take as inputs somewhat simplified interactions [13]. In addition, the use of wave functions that scale polynomially with the number of nucleons exacerbates the AFDMC fermion sign problem for $A > 16$ nuclei.

Algorithms that take advantage of noisy intermediate-scale quantum devices are potentially groundbreaking alternatives, and their capabilities have already been demonstrated on prototypal nuclear problems [14–16]. Another class of approaches being actively explored is based on machine learning (ML) techniques. These techniques typically rely on the ability of artificial neural networks (ANNs) to compactly represent complex high-

dimensional functions, as already leveraged in several domains of physics [17]. For many-body quantum applications, neural-network-based variational representations have been introduced in Ref. [18], and have found applications as a tool to study ground state and dynamics of several interacting lattice quantum systems [19–27]. In a series of recent works [28–30] deep neural networks have been further developed to tackle ab-initio chemistry problems within variational Monte Carlo, often resulting in accuracy improvements over existing variational approaches traditionally used to describe correlated molecules.

While applications of ML approaches to the many-body problem in condensed matter, quantum chemistry, and quantum information have been proliferating in the past few years, the adoption in low-energy nuclear theory is still in its infancy [31, 32]. Pioneering work in the field [33] has provided a proof-of-principle application of ANN to solve the Schrödinger equation of the deuteron. Extending the non-stochastic approach of Ref. [33] to larger nuclei however presents an intrinsically exponentially scaling challenge. In this work, we expand the domain of applicability of ANN-based representations of the wave function and compute ground-state properties of $A \leq 4$ nuclei as they emerge from a leading-order pionless effective field theory (EFT) Hamiltonian, containing consistent two- and three-body potentials. Specifically, we develop a nuclear VMC algorithm that capitalizes on the stochastic-reconfiguration algorithm [34] to efficiently train the parameters of the ANN. We benchmark our results against a more conventional parametrization of the variational wave function in terms of two- and three-body Jastrow functions, and virtually-exact GFMC cal-

culations.

Hamiltonian.— We employ nuclear Hamiltonians derived within pionless EFT, which is based on the tenet that the typical momentum of nucleons in nuclei is much smaller than the pion mass m_π [3, 35]. Under this assumption, largely justified for studying the structure and long-range properties of $A \leq 4$ nuclei, pion exchanges are unresolved contact interactions and nucleons are the only relevant degrees of freedom. The singularities of the contact terms are controlled introducing a Gaussian regulator that suppresses transferred momenta above the ultraviolet cutoff Λ . This regulator choice directly leads to a Gaussian radial dependence of the potential, which is local in coordinate [36, 37]. The leading-order (LO) Hamiltonian reads

$$H_{LO} = - \sum_i \frac{\vec{\nabla}_i^2}{2m_N} + \sum_{i < j} (C_1 + C_2 \vec{\sigma}_i \cdot \vec{\sigma}_j) e^{-r_{ij}^2 \Lambda^2 / 4} + D_0 \sum_{i < j < k} \sum_{\text{cyc}} e^{-(r_{ik}^2 + r_{ij}^2) \Lambda^2 / 4}, \quad (1)$$

where m_N is the mass of the nucleon, $\vec{\sigma}_i$ is the Pauli matrix acting on nucleon i , and $\sum_{\text{cyc}}$ stands for the cyclic permutation of i, j , and k .

Following Ref. [38], the low-energy constants C_1 and C_2 are fit to the deuteron binding energy and to the neutron-neutron scattering length. In Eq. (1) we picked the operator basis 1 and $\vec{\sigma}_i \cdot \vec{\sigma}_j$, but this choice can be replaced by any other form equivalent under Fierz transformations in SU(2). Solving $A \geq 3$ nuclei with purely attractive two-nucleon potentials leads to the “Thomas collapse” [39], which can be avoided promoting a contact three-nucleon force to LO [40]. The values of the LECs’ adopted in this work can be found in Ref. [38]; since $C_1(\Lambda)$ is much larger than $C_2(\Lambda)$, the LO Hamiltonian has an approximate SU(4) symmetry.

Variational wave function.— A fundamental ingredient of the VMC method is the choice of a suitable variational wave function Ψ_V , whose parameters are found exploiting the variational principle

$$\frac{\langle \Psi_V | H | \Psi_V \rangle}{\langle \Psi_V | \Psi_V \rangle} = E_V \geq E_0 \quad (2)$$

where E_0 is exact the ground-state energy: $H|\Psi_0\rangle = E_0|\Psi_0\rangle$. Metropolis Monte Carlo sampling is used to evaluate E_T through a $3A$ -dimensional integral in coordinate space. The vast majority of nuclear correlations induced by the LO pionless nuclear Hamiltonian of Eq. (1), which is approximately SU(4) symmetric, are spin-isospin independent. To accurately model them, we introduce an ANN representation of the variational wave function

$$|\Psi_V^{\text{ANN}}\rangle = e^{\mathcal{U}(\mathbf{r}_1, \dots, \mathbf{r}_A)} |\Phi\rangle \quad (3)$$

where $|\Phi\rangle$ is a suitable mean-field state, and $\mathcal{U}(\mathbf{r}_1, \dots, \mathbf{r}_A)$ is a real-valued function. For the s-shell nuclei considered in this work, the mean-field part can be taken to be

$|\Phi_{2\text{H}}\rangle = \mathcal{A}|\uparrow_p \uparrow_n\rangle$, $|\Phi_{3\text{He}}\rangle = \mathcal{A}|\uparrow_p \downarrow_p \uparrow_n\rangle$, and $|\Phi_{4\text{He}}\rangle = \mathcal{A}|\uparrow_p \downarrow_p \uparrow_n \downarrow_n\rangle$, with $\mathcal{A}$ being the anti-symmetrization operator [41].

The correlating factor $\mathcal{U}(\mathbf{r}_1, \dots, \mathbf{r}_A)$ is parameterized with an ANN comprised of four fully connected layers. The input layer has $3A$ nodes, corresponding to the intrinsic spatial coordinates of the nucleon, defined as $\bar{\mathbf{r}}_i = \mathbf{r}_i - \mathbf{R}_{\text{CM}}$, where $\mathbf{R}_{\text{CM}}$ is the center of mass coordinate. This procedure automatically removes spurious center of mass contributions from all observables [42]. The two intermediate layers have 128 nodes each, while the output layer is one-dimensional. The network has a total of 18304 trainable parameters. Since the kinetic energy requires evaluating the derivatives of Ψ_T with respect to $\mathbf{r}_i$, we use the differentiable softplus activation functions [43]. The parameters of the network are randomly initialized, hence during the Metropolis walk the nucleons can in principle drift away from $\mathbf{R}_{\text{CM}}$. To control this pathological behavior, a Gaussian function is added to effectively confine the nucleons within a finite volume $\mathcal{U}(\mathbf{r}_1, \dots, \mathbf{r}_A) \rightarrow \mathcal{U}(\mathbf{r}_1, \dots, \mathbf{r}_A) - \alpha \sum_i \bar{\mathbf{r}}_i^2$ where we take $\alpha = 0.05$.

The choice of correcting a mean-field state $|\Phi\rangle$ with a flexible ANN correlator factor $\mathcal{U}(\mathbf{r}_1, \dots, \mathbf{r}_A)$ is similar in spirit to neural-network correlators introduced recently in condensed-matter [19, 27] and chemistry applications [29]. An appealing feature of the ANN ansatz is that it is more general than the more conventional product of two- and three-body spin-independent Jastrow functions

$$|\Psi_V^J\rangle = \prod_{i < j < k} \left(1 - \sum_{\text{cyc}} u(r_{ij})u(r_{jk})\right) \prod_{i < j} f(r_{ij}) |\Phi\rangle, \quad (4)$$

which is commonly used for nuclear Hamiltonians that do not contain tensor and spin-orbit terms [37].

Analogously to standard VMC calculations, as well as ML applications, the optimal set of weights and biases of the ANN is found minimizing a suitable cost function. Specifically, we exploit the variational principle of Eq. (2)

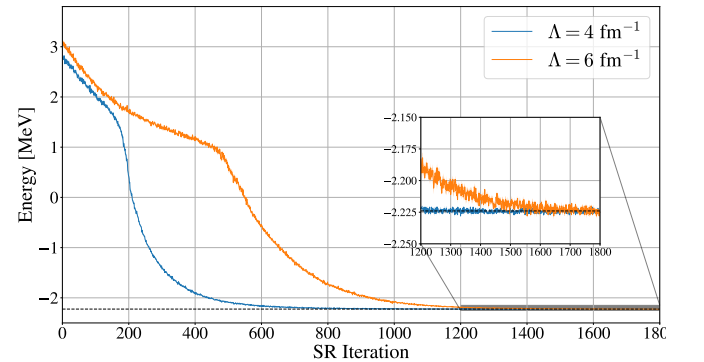


FIG. 1. Convergence pattern of the ${}^2\text{H}$ variational energy for $\Lambda = 4 \text{ fm}^{-1}$ and $\Lambda = 6 \text{ fm}^{-1}$ as a function of the number of optimization steps of the SR algorithm. The dashed line denotes the asymptotic value.

and minimize the expectation value of the energy. The gradient components $G_i = \partial_i E(p)$ of the energy with respect to the variational parameters p_i read

$$G_i = 2 \left(\frac{\langle \partial_i \Psi_V | H | \Psi_V \rangle}{\langle \Psi_V | \Psi_V \rangle} - E_V \frac{\langle \partial_i \Psi_V | \Psi_V \rangle}{\langle \Psi_V | \Psi_V \rangle} \right) \quad (5)$$

and can be efficiently estimated through Monte Carlo sampling. While stochastic gradient descent can be readily used to compute parameters updates, for VMC applications it has been found that using a preconditioner based on the Quantum Fisher Information

$$S_{ij} = \frac{\langle \partial_i \Psi_V | \partial_j \Psi_V \rangle}{\langle \Psi_V | \Psi_V \rangle} - \frac{\langle \partial_i \Psi_V | \Psi_V \rangle \langle \Psi_V | \partial_j \Psi_V \rangle}{\langle \Psi_V | \Psi_V \rangle \langle \Psi_V | \Psi_V \rangle}, \quad (6)$$

is significantly more efficient. During the optimization, then parameters at step s are updated as $p^{s+1} = p^s - \eta(S + \Lambda)^{-1}G$, where η is the learning rate and Λ is a small positive diagonal matrix that is added to stabilize the method. This approach, known as the stochastic-reconfiguration (SR) algorithm [34, 44] is equivalent to performing imaginary-time evolution in the variational manifold and it is in turn related to the Natural Gradient descent method [45] in unsupervised learning.

Figure 1 displays the convergence pattern as function of the optimization step of the ^2H energy for the LO pionless EFT Hamiltonians with $\Lambda = 4 \text{ fm}^{-1}$ and $\Lambda = 6 \text{ fm}^{-1}$. The softer cutoff exhibit a faster convergence, reaching its asymptotic value after about 1100 iterations, while the stiffer regulator takes longer to converge: about 1500 optimization steps. These results have been obtained with a relatively small learning rate, $\eta = 0.001$, which has proven to yield robust convergence patterns for all the nuclei and regulator choices that we have analyzed. Using less conservative learning rates requires performing heuristic tests on the parameter change, similar to the ones introduced in Ref. [37, 42] for regularizing the linear optimization method [46].

The general ML framework *Pytorch* [47] is adopted to construct the network and efficiently compute the derivatives with respect to both the network parameters and the position of the nucleons. We note that the automatic differentiation capabilities and GPU-native framework of *Pytorch* imply efficient scaling of this technique, though the imperative nature of operations limit the scalability of Jacobian calculations in *Pytorch*. A just-in-time (JIT) compiled (Jax [48]) or graph-optimized framework (Tensorflow [49]) shows much better preliminary performance on state-of-the-art computing hardware.

Results and discussion. - We analyze the accuracy of the ANN wave function ansatz by computing the ground-state energies of ^2H , ^3H , and ^4He as they emerge from the LO pionless EFT Hamiltonians with regulators $\Lambda = 4 \text{ fm}^{-1}$ and $\Lambda = 6 \text{ fm}^{-1}$. In Table I we benchmark the ANN representation of Ψ_T (VMC-ANN) against conventional VMC calculations carried out using a spline

	Λ	VMC-ANN	VMC-JS	GFMC	GFMC _c
^2H	4 fm^{-1}	-2.224(1)	-2.223(1)	-2.224(1)	-
	6 fm^{-1}	-2.224(4)	-2.220(1)	-2.225(1)	-
^3H	4 fm^{-1}	-7.81(1)	-7.80(1)	-8.38(2)	-7.82(1)
	6 fm^{-1}	-7.79(3)	-7.74(1)	-8.38(2)	-7.81(1)
^4He	4 fm^{-1}	-22.76(3)	-22.54(1)	-23.62(3)	-22.77(2)
	6 fm^{-1}	-23.99(6)	-23.44(2)	-25.06(3)	-24.10(2)

TABLE I. Ground-state energy of the ^2H , ^3H , and ^4He for the LO pionless-EFT Hamiltonian for $\Lambda = 4 \text{ fm}^{-1}$ and $\Lambda = 6 \text{ fm}^{-1}$. Numbers in parentheses indicate the statistical errors on the last digit.

parametrization for the Jastrow functions [37] (VMC-JS), and virtually-exact GFMC results.

As far as ^2H is concerned, the three methods provide fully compatible energies, within statistical errors, showing the flexibility of the ANN to accurately represent the ground-state wave function of the deuteron, consistently with the findings of Ref. [33]. Note that, since the LO pionless EFT Hamiltonian does not contain tensor or spin-orbit terms, the VMC-JS ansatz is exact. The perfect agreement with the experimental value is not surprising, as the potential has been fit to the deuteron binding energy using numerically-exact few-body methods [36].

The VMC-ANN and VMC-JS energies of ^3H energies essentially coincide, as the variational wave function of Eq. (4), containing two- and three-body correlations, is flexible enough to model the spin-independent correlations relevant for systems with up to $A = 3$ nucleons. On the other hand, the GFMC results are moderately more bound, by about 0.6 MeV, for both $\Lambda = 4 \text{ fm}^{-1}$ and $\Lambda = 6 \text{ fm}^{-1}$. This difference is due to spin-dependent correlations that are automatically generated by the GFMC imaginary-time propagation, but are neglected in both the VMC-JS and VMC-ANN calculations. In the latter cases, the expectation value of $\vec{\sigma}_i \cdot \vec{\sigma}_j$ is exactly equal to -1, and the NN potential of Eq. (1) is equivalent to using $\tilde{v}_c(r_{ij}) = v_c(r_{ij}) - v_\sigma(r_{ij})$. To better quantify the amount of spin-independent correlations entailed in the ANN, we have carried out GFMC calculations using this purely central interaction. The corresponding GFMC_c energies are compatible with the ANN results within statistical uncertainties, proving the high-accuracy of the ANN wave-function ansatz. A similar pattern emerges for the ^4He nucleus, although in this case the ANN wave functions outperform the JS ones – the energy is improved by about 0.2 MeV and 0.5 MeV for $\Lambda = 4 \text{ fm}^{-1}$ and $\Lambda = 6 \text{ fm}^{-1}$, respectively. Spin-dependent correlations lower the GFMC energies by about 1 MeV with respect to the VMC-ANN ones for both values of the regulator. On the other hand, VMC-ANN and GFMC_c results agree for $\Lambda = 4 \text{ fm}^{-1}$, while for $\Lambda = 6 \text{ fm}^{-1}$ GFMC_c provides $\simeq 0.1$ MeV of additional binding. We ascribe this small difference to the difficulties inherent in optimizing the wave function when stiff regulators are employed, in line

with the convergence pattern of the ^2H energy displayed in Fig. 1.

To further elucidate the quality of the ANN wave function we consider the point-nucleon density

$$\rho_N(r) = \frac{1}{4\pi r^2} \langle \Psi_V | \sum_i \delta(r - |\mathbf{r}_i^{\text{int}}|) | \Psi_V \rangle, \quad (7)$$

which is of interest in a variety of experimental settings [50, 51]. In the upper, medium, and lower panels of Fig. 2 we display $\rho_N(r)$ of ^2H , ^3H , and ^4He as obtained from VMC-ANN and GFMC calculations that use as input the LO pionless-EFT Hamiltonian with $\Lambda = 4 \text{ fm}^{-1}$. There is an excellent agreement between the two methods, which further corroborates the representative power

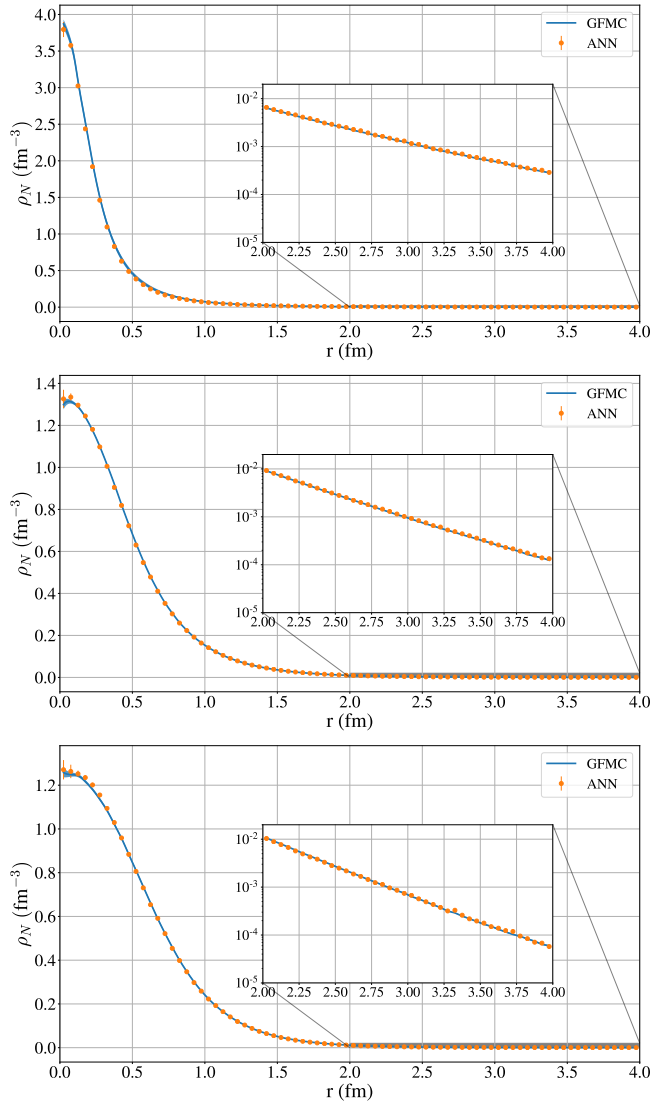


FIG. 2. Point-nucleon densities of ^2H (upper panel), ^3H (middle panel), and ^4He (lower panel) for the LO pionless-EFT Hamiltonian with $\Lambda = 4 \text{ fm}^{-1}$. The solid points and the shaded area represent the VMC-ANN and GFMC results, respectively.

of the ANN ansatz for the wave functions of $A \leq 4$ nuclei. The VMC-ANN and GFMC densities overlap both at short distances and in the slowly-decaying asymptotic exponential tails, highlighted in the insets of Fig. 2. It has to be emphasised that the ANN learns how to compensate for the original Gaussian confining function and reproduce the correct exponential falls off of the nuclear wave function, which is notoriously delicate to obtain within nuclear methods that rely on harmonic-oscillator basis expansions [52, 53].

Conclusions – In this work we have carried out proof-of-principle calculations that demonstrate the capability of ANNs to represent the variational state of $A \leq 4$ nuclei. Capitalizing on the stochastic-reconfiguration algorithm, we trained the network to solve the Schrödinger equation of a LO pionless-EFT Hamiltonian that contains two- and three-nucleon potentials characterized by highly non-perturbative short-range components. Our variational Monte Carlo calculations demonstrate that the ANN variational wave function outperforms the routinely employed two- and three-body Jastrow parametrization of the correlation function. The small differences with respect to virtually-exact GFMC results are due to missing spin-dependent correlations, as shown by GFMC calculations carried out using spin-independent interactions. The single-particle densities obtained with ANN wave functions are also in excellent agreement with GFMC results, both at short distances and in the slowly-decaying exponential tails, which are notoriously difficult to reproduce. For the s-shell nuclei that we have considered in this work, and with our analysis limited to spin-isospin independent correlations, the training naturally converges to permutation-invariant wave functions. The extension to spin-isospin dependent correlation requires developing complex-valued representations of the variational wave function that are anti-symmetric under the simultaneous exchange of the position and spin-isospin coordinates of any two nucleons. While this aspect is left for future research activity, we anticipate that the adoption of strategies already successfully adopted in fermionic many-body problems [28–30] will be beneficial also for applications in nuclear systems.

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